

3.3 Halogen Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.3 Halogen Compounds
Difficulty	Medium

Time allowed: 100

Score: /76

Percentage: /100

Question 1a

The molecular formula C_3H_6 represents the compounds propene and cyclopropane are shown in Fig. 1.1.



Fig. 1.1

What is the H-C-H bond angle at the terminal $=CH_2$ group in propene?

[1 mark]

Question 1b

Under suitable conditions, propene and cyclopropane each react with chlorine.

i)

With propene, 1,2-dichloropropane, $CH_3CHClCH_2Cl$ is formed.

State fully what type of reaction this is.

[1]

ii)

When cyclopropane reacts with chlorine, **three** different compounds with the molecular formula $C_3H_4Cl_2$ can be formed.

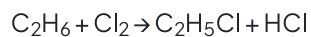
Draw displayed structures of each of **these** three compounds.

[3]

[4 marks]

Question 1c

Ethane reacts with chlorine according to the following equation.



i)

State the conditions needed for this reaction.

[1]

ii)

State the type of reaction occurring here.

[1]

[2 marks]

Question 1d

i)

One of the steps during the reaction in (c) is the following process.

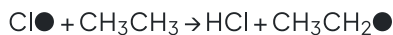


Table 1.1

Bond	Bond enthalpy / kJ mol^{-1}
C-H	410
H-Cl	431
H-I	299

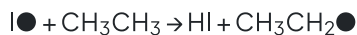
Use the data in Table 1.1 to calculate the enthalpy change, ΔH , of this step.

Show your working.

$$\Delta H = \dots\dots\dots \text{kJ mol}^{-1} \quad [1]$$

ii)

Use the data in Table 1.1 to calculate the enthalpy change, ΔH , of the similar reaction:



Show your working.

$$\Delta H = \dots\dots\dots \text{kJ mol}^{-1} \quad [1]$$

iii)

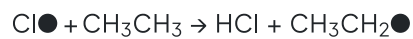
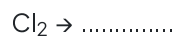
Hence suggest why it is not possible to make iodoethane by reacting together iodine and ethane.

[1]

[3 marks]

Question 1e

Complete the following equations of some possible steps in the formation of chloroethane.

**[3 marks]**

Question 2a

This question concerns halogenoalkanes.

1-chloropropane can react to form organic products as shown in the reaction scheme in Fig. 2.1.

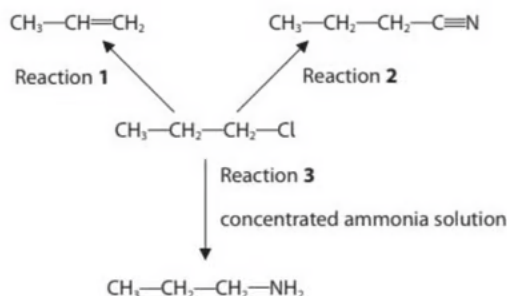


Fig. 2.1

- i)
State the reagent and conditions used in Reaction 1.

[2]

- ii)
Identify a suitable reagent for Reaction 2 and include a reason why this is a particularly useful type of reaction in organic chemistry.

[2]

Reagent

Reason

- iii)
State the conditions which are used in Reaction 3.

[2]

- iv)
Write the structural formula of the product that will be formed if 1-chloropropane is refluxed with aqueous sodium hydroxide solution.

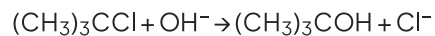
[1]

[7 marks]

Question 2b

2-chloro-2-methylpropane can be reacted with aqueous alkali to form 2-methylpropan-2-ol.

The equation for this reaction is



The graph in Fig 2.2 shows how the concentration of 2-chloro-2-methylpropane changes with time during an investigation of this reaction.

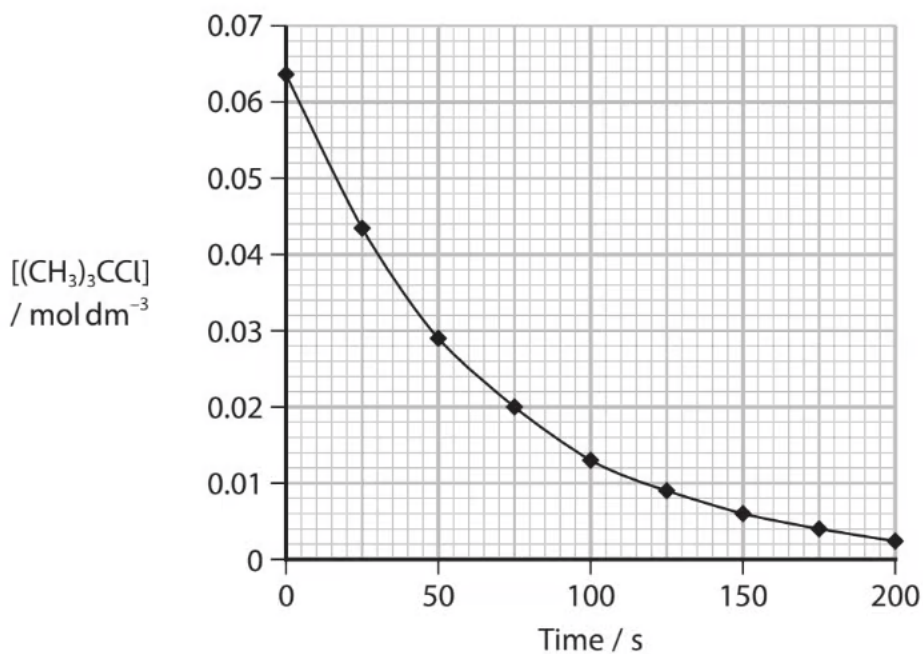


Fig. 2.2

Calculate the rate of reaction at 50 s. Show your working on the graph.
Include units with your final answer.

Rate of reaction at 50 s =

[3 marks]

Question 2c

What is the classification of the mechanism for the reaction of 2-chloro-2-methylpropane with aqueous alkali?

[1 mark]

Question 2d

The letters **X**, **Y** and **Z** refer to three different halogenoalkanes:

X	1-bromobutane
Y	1-iodobutane
Z	1-chlorobutane

1 cm³ of each of these halogenoalkanes was added to separate test tubes containing 5 cm³ of ethanol and 5 cm³ of aqueous silver nitrate solution in a water bath at 50 °C.

- i)
State the visible change in the reaction of an ethanol/silver nitrate solution with halogenoalkane **X**.
Include the **formula** of the compound responsible for this observation.

[2]

- ii)
The three halogenoalkanes were placed in order of **decreasing** rate of reaction.
State the order, starting with the halogenoalkane that reacted the fastest

[1]

[3 marks]

Question 2e

2-bromo-2-methylpropane can be prepared from the addition of hydrogen bromide, HBr, to 2-methylpropene.

Draw the mechanism for this reaction.
Include curly arrows, and any relevant dipoles and lone pairs.

[4 marks]

Question 3a

One method of making 1-bromobutane in the laboratory is described below.

- Step 1** Place 35 g of powdered potassium bromide, 30 cm³ of water, and 25 cm³ of butan-1-ol, in a 250 cm³ two necked flask fitted with a tap funnel and reflux condenser.
- Step 2** Concentrated sulfuric acid (25 cm³) is then placed in the tap funnel and added drop by drop to the reagents in the flask, keeping the contents well shaken and cooled occasionally in an ice-water bath.

The overall reaction may be considered to take place in two stages. In the first stage, the inorganic reagents react together to form HBr. In the second stage, the organic reagent reacts with the HBr that is formed in the first stage.

Write an equation for each of these stages.

Stage 1

Stage 2

[2 marks]

Question 3b

1-bromobutane can be prepared using different reagents and conditions.

State **two** different ways that 1-bromobutane can be prepared.

In each case, include the reagents and conditions that are necessary

[2 marks]

Question 3c

i)
Complete the following reaction scheme shown in Fig. 3.1 which starts with 1-bromobutane.

In each empty box, write the structural formula of the organic compound that would be formed.

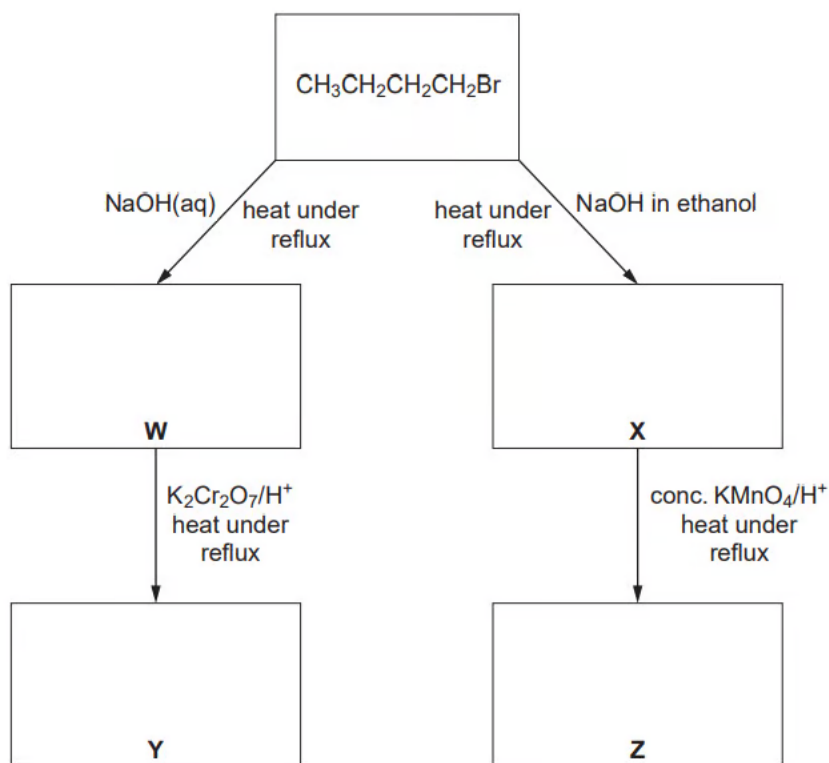


Fig. 3.1

ii)
State the name of the type of reaction that occurs when **X** is produced from 1-bromobutane.

[5 marks]**Question 3d**

Compound **W** can also be formed from 1-bromobutane by heating it under reflux with aqueous silver nitrate. This causes the hydrolysis of 1-bromobutane.

Suggest why the rate of the formation of compound **W** is slower using this method than heating under reflux with aqueous sodium hydroxide.

[2 marks]**Question 4a**

This is a question about the hydrolysis of halogenoalkanes.

Devise an experiment, giving outline details only, that would enable the relative rates of hydrolysis of halogenoalkanes to be compared.

[5 marks]

Question 4b

Explain the trend in the rates of hydrolysis of 1-chlorobutane, 1-bromobutane and 1-iodobutane.

[2 marks]

Question 4c

The product of the hydrolysis of 2-bromobutane is butan-2-ol. Both molecules are chiral.

State what is meant by the term **chiral**, using three-dimensional diagrams of the enantiomers of butan-2-ol to illustrate your answer.

[3 marks]

Question 4d

Compare and contrast the mechanism of hydrolysis, using aqueous sodium hydroxide, of the primary halogenoalkane, RCH_2X , with that of the tertiary halogenoalkane, R_3CX .

Include diagrams of any intermediate or transition state.

Curly arrows are not required.

[6 marks]**Question 5a**

Give the structures of the four structural isomers of $\text{C}_4\text{H}_9\text{Br}$ **and** identify each as primary, secondary or tertiary.

[4 marks]

Question 5b

Name the isomer of C_4H_9Br that contains a chiral centre and draw the three-dimensional structures of the two optical isomers.

name

structures

--	--

[3 marks]

Question 5c

Aqueous silver nitrate solution was added to separate tubes containing chloroethane, bromoethane and iodoethane. The tubes were heated in a water bath.

A yellow precipitate appeared first in the tube containing iodoethane, followed by a cream precipitate in the tube containing bromoethane and finally a white precipitate appeared in the tube containing chloroethane.

Explain these observations.

[2 marks]

Question 5d

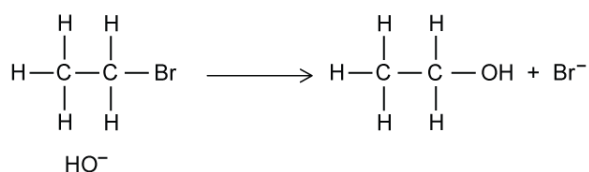
i)

Give the full name of the mechanism for the reaction between aqueous sodium hydroxide and bromoethane.

[2]

ii)

Complete the diagram below to represent this mechanism. Include all necessary curly arrows, partial charges and lone pairs.



[2]

[4 marks]

Question 5e

Primary and tertiary halogenoalkanes have different reaction mechanisms when they react with aqueous sodium hydroxide.

i)

State the name that is given to the reaction mechanism when aqueous sodium hydroxide is reacted with the following types of halogenoalkane.

primary halogenoalkane

tertiary halogenoalkane

[2]

ii)

Tertiary halogenoalkanes form a tertiary carbocation as an intermediate during this reaction. Explain why tertiary carbocations are more stable than primary carbocations.

[3]

[5 marks]

Model Answers: Medium

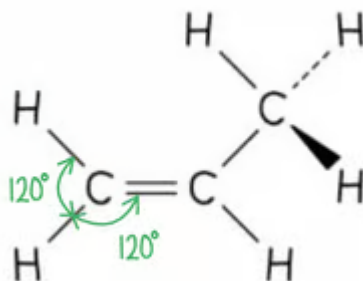
1a

a) The H-C-H bond angle at the terminal =CH₂ group in propene is:

- 120°; [1 mark]

[Total: 1 mark]

- There are 3 bonded regions around the terminal carbon, 2 C-H bonds and 1 C=C bond
- The repulsion between single bonds and double bonds is treated the same as the repulsion between single bonds and single bonds
- The three bonded regions will therefore repel each other equally, giving a bond angle of 120° which forms a trigonal planar shape around the terminal C atom
- An angle between 117° and 120° would be accepted



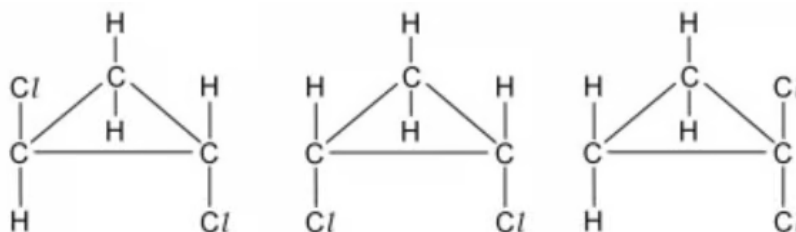
1b

b)

i) The type of reaction is:

- Electrophilic addition; [1 mark]

ii) The displayed structures of each of these **three** compounds are:



- [1 mark] for each correct structure formed

[Total: 4 marks]

- The δ^+ Cl in a Cl_2 molecule will act as an **electrophile** and the chlorine will **add** across the double bond, therefore the reaction is electrophilic addition
- The three isomers will have 2 Cl atoms in different places - refer back to the original diagram of cyclopropane to deduce the different combinations:
 - One Cl on each C atom with one Cl atom above and one below the plane of C atoms
 - One Cl on each C atom with both either above or below the plane of C atoms
 - Two Cl atoms on one C atom with one above and one below the plane of C atoms

1c

c)

i) The conditions needed for this reaction is:

- UV / ultraviolet light; [1 mark]

ii) The type of reaction occurring here is:

- (Free) radical substitution; [1 mark]

[Total: 2 marks]

- You could also state 300° for the conditions
- This is one of three methods by which halogenoalkanes can be formed, along with the electrophilic addition of alkenes, as seen in the previous question and the substitution of alcohols
- This question shows that you need to know all of the different methods

1d

d)

i) The enthalpy change, ΔH , of this step is:

- $\Delta H = \text{bond energy (C-H)} - \text{bond energy (H-Cl)} = 410 - 431 = -21 \text{ kJ mol}^{-1}$; [1 mark]

ii) The enthalpy change, ΔH , of the similar reaction is:

- $\Delta H = \text{bond energy (C-H)} - \text{bond energy (H-I)} = 410 - 299 = +111 \text{ kJ mol}^{-1}$; [1 mark]

iii) It is not possible to make iodoethane by reacting together iodine and ethane because:

- The reaction with iodine is endothermic

OR

ΔH is positive

OR

The reaction with iodine requires energy; [1 mark]

[Total: 3 marks]

- The enthalpy change of a reaction is calculated by:
 - enthalpy change for bonds broken + enthalpy change for bonds formed
- The enthalpy change for bonds broken is the sum of the bond energies for all of the bonds broken
- The enthalpy change for bonds formed is the **negative** of the sum of the bond energies for all of the bonds formed
 - It is the negative of this value as energy is released when bonds are formed, i.e. it is an exothermic process
- So the enthalpy change of the reaction can also be calculated by:
 - sum of the bond energies for bonds broken - sum of the bond energies for bonds formed
- In the first reaction, 1 C-H bond is broken and 1 H-Cl bond is formed
- In the second reaction, 1 C-H bond is broken and 1 H-I bond is formed

1e

e) The completed equations of some possible steps in the formation of chloroethane

are:

- $\text{Cl}_2 \rightarrow 2\text{Cl}\bullet$; [1 mark]
- $\text{CH}_3\text{CH}_2\bullet + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{Cl}\bullet$; [1 mark]
- $\text{CH}_3\text{CH}_2\bullet + \text{Cl}\bullet \rightarrow \text{CH}_3\text{CH}_2\text{Cl}$; [1 mark]

[Total: 3 marks]

- The first step shows initiation where the bond in a chlorine molecule is broken homolytically to form two chlorine radicals
- The second step shows propagation where one radical reacts and another radical is produced
 - This step shows two products being formed, so it cannot be a termination step with another radical, as only one product would be formed if this were the case
- The third step shows termination as the end product does not contain a free radical, this occurs when two radicals react together

2a

a)

i) The reagent and conditions used in Reaction 1 are:

- Reagent: (concentrated) NaOH / KOH; [1 mark]
- Conditions: ethanol (solvent)
AND
Heat / warm; [1 mark]

ii) A suitable reagent, and reason, for Reaction 2 is:

- Reagent: KCN / NaCN / potassium cyanide / sodium cyanide; [1 mark]
- Reason: it increases the number of carbon atoms in the carbon chain / length of carbon chain; [1 mark]

iii) The conditions that are used in Reaction 3 are:

- Heating; [1 mark]
- Under pressure; [1 mark]

iv) The structural formula of the product that will be formed if 1-chloropropane is refluxed with an aqueous potassium hydroxide solution is:

- $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$; [1 mark]

[Total: 7 marks]

- In part (i), the second mark is dependent on getting the first mark, except for a 'near miss', e.g. you said OH^- instead of NaOH or KOH
 - Instead of saying heat or warm, you could give reflux as the condition
- In part (ii) nitriles can be formed from halogenoalkanes by heating under reflux with an ethanolic solution of potassium (or sodium) cyanide
 - This is an important reaction as it can be used to lengthen the carbon chain
 - You do not need to give the conditions for this reaction so any reference to these will be ignored
- When giving the structural formula in part (iv), you would be awarded the mark if you gave the displayed or skeletal formula but not for $\text{C}_3\text{H}_7\text{OH}$
- This question shows the need to learn the different reactions together with the necessary reagents and conditions, for the different organic species in the syllabus

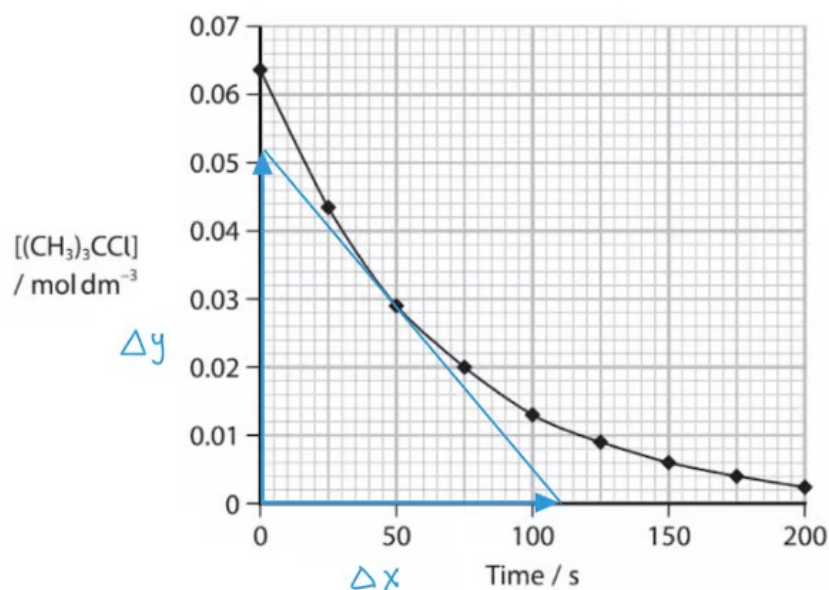
2b

b) The rate of reaction at 50 s is:

- Tangent drawn on graph at 50 s; [1 mark]
- Calculation of rate; [1 mark]
- Units; [1 mark]

[Total: 3 marks]

- Draw a tangent to the curve at 50 seconds



- Calculate the gradient using $\frac{\Delta y}{\Delta x} = -\frac{0.052}{110} = (-)4 \times 10^{-4} \text{ mol dm}^3 / \text{s}$
- Answers in the range of $(-)4.0 - (-)6.0 \times 10^{-4}$
- Units of concentration can also be written as $\text{mol dm}^{-3} \text{ s}^{-1}$ or $\text{mol dm}^{-3} \text{ per s}$

2c

c) The classification of the mechanism for the hydrolysis of 2-chloro-2-methylpropane by water is:

- Nucleophilic substitution; [1 mark]

[Total: 1 mark]

- 2-chloro-2-methylpropane undergoes nucleophilic substitution due to the presence of polar C-Cl bonds
- The hydroxide ion, a nucleophile, can attack the carbon atom in the C-Cl bond, resulting in the substitution of -Cl for -OH and an alcohol being formed

2d

d)

i) The visible change in the reaction of an ethanol/silver nitrate solution with halogenoalkane **X** is:

- Cream precipitate; [1 mark]
- AgBr; [1 mark]

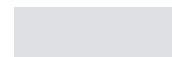
ii) The three halogenoalkanes in order of **decreasing** rate of reaction are:

- Y,X,Z

OR

1-iodobutane, 1-bromobutane, 1-chlorobutane; [1 mark]

[Total: 3 marks]



- The hydroxide ion, a nucleophile, can attack the carbon atom in the C-Cl bond, resulting in the substitution of -Cl for -OH and an alcohol being formed

2d

d)

i) The visible change in the reaction of an ethanol/silver nitrate solution with halogenoalkane **X** is:

- Cream precipitate; [1 mark]
- AgBr; [1 mark]

ii) The three halogenoalkanes in order of **decreasing** rate of reaction are:

- Y,X,Z

OR

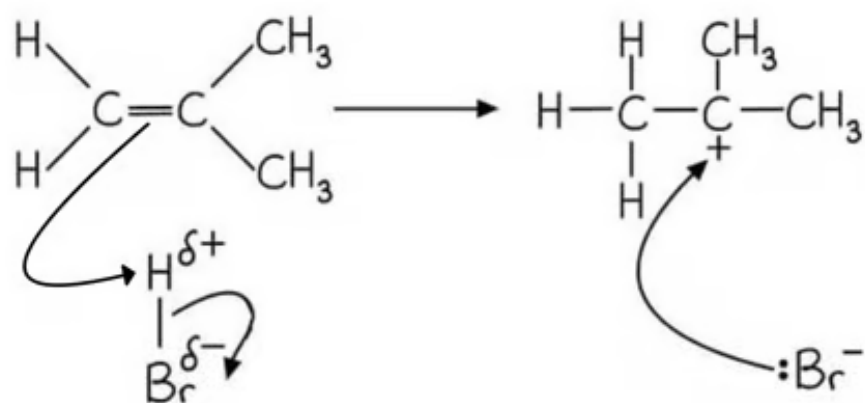
1-iodobutane, 1-bromobutane, 1-chlorobutane; [1 mark]

[Total: 3 marks]

- You will also be awarded the mark if you give the precipitate colour as off-white / or very pale yellow (but not just yellow)
- You could also write ppt / ppte / solid or crystals as opposed to precipitate
- The following reaction occurs to form the precipitate:
 - $\text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$
- The order of decreasing reactivity for halogenoalkanes is iodoalkanes > bromoalkanes > chloroalkanes

2e

e) A mechanism for the addition of hydrogen bromide, HBr, to 2-methylpropene to form 2-bromo-2-methylpropane is:



- Curly arrow from double bond to H of HBr
AND
Correct structure of 2-methylpropene; [1 mark]
- Curly arrow from H-Br bond to Br atom
AND
Correct dipole on HBr molecule; [1 mark]
- Intermediate with + on correct Carbon
AND
 Br^- ; [1 mark]
- Lone pair on Br^-
AND
Curly arrow from lone pair to C^+ ; [1 mark]

[Total: 4 marks]

- 2-methylpropene will undergo electrophilic addition to form 2-bromo-2-methylpropane
 - Read the mark scheme carefully to see exactly where you get the marks for this mechanism
 - You do not need to draw the final product, this is not a marking point
 - If you get the structure of 2-methylpropene incorrect you will lose the first mark but can achieve the rest for the correct mechanism on an incorrect alkene
 - If + is written on an incorrect carbon you would lose mark three but can still achieve mark four

3a

a) An equation for each of these stages is:

- Stage 1: $\text{KBr} + \text{H}_2\text{SO}_4 \rightarrow \text{KHSO}_4 + \text{HBr}$

OR



- Stage 2: $\text{C}_4\text{H}_9\text{OH} + \text{HBr} \rightarrow \text{C}_4\text{H}_9\text{Br} + \text{H}_2\text{O}; [1 \text{ mark}]$

[Total: 2 marks]

- **Careful:** This question is not asking you to give an equation for each step
- The possible reagents are: potassium bromide, butan-1-ol and concentrated sulfuric acid
- The question tells you that the first stage of the reaction produces HBr from inorganic reagents, so this cannot involve butan-1-ol
- Butan-1-ol is a reactant in the 2nd stage along with HBr to form 1-bromobutane

3b

b) **Two** different ways that 1-bromobutane can be prepared are:

Any **two** from:

- Br_2 / bromine **AND** butane **AND** UV light; [1 mark]
- Butene **AND** HBr; [1 mark]
- Butan-1-ol **AND** PCl_3 **AND** heat; [1 mark]
- Butan-1-ol **AND** PCl_5 (at room temperature); [1 mark]
- Butan-1-ol **AND** SOCl_2 ; [1 mark]

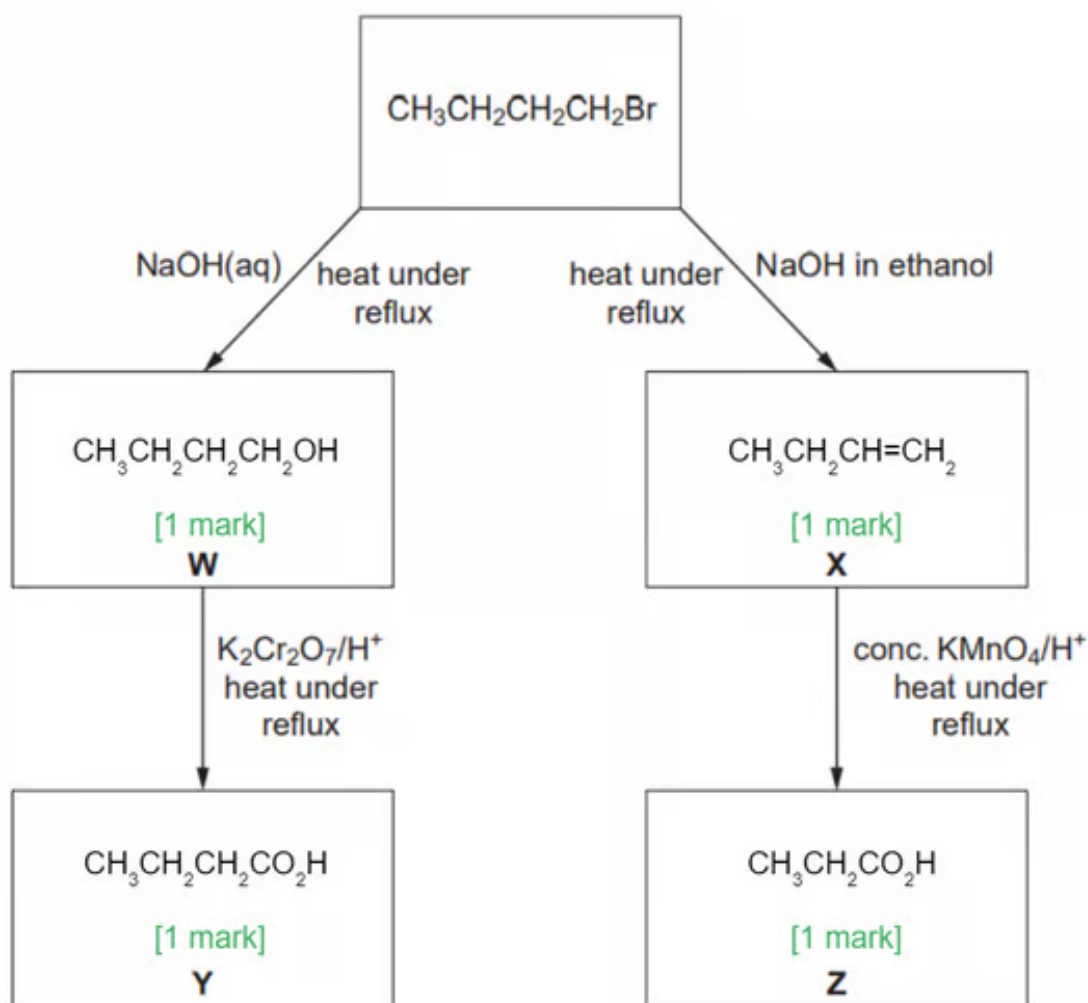
[Total: 2 marks]

- There are several acceptable methods, just make sure you state any reaction conditions as necessary
- Butan-1-ol and HBr / $\text{KBr} + \text{H}_2\text{SO}_4$ / H_3PO_4 is not accepted as this is the method that is described in part (a)

3c

c)

i) The completed reaction scheme which starts with 1-bromobutane is:



ii) The type of reaction that occurs when **X** is produced from 1-bromobutane is:

- Elimination; [1 mark]

[Total: 5 marks]

- You need to know the different products that halogenoalkanes form when reacted with NaOH, which differ depending on the conditions used
- When halogenoalkanes are reacted with aqueous sodium hydroxide, a nucleophilic substitution reaction occurs to form an alcohol
- When halogenoalkanes are heated with ethanolic sodium hydroxide, an elimination reaction occurs forming an alkene, sodium halide and water
- When an alcohol is heated under reflux with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, oxidation occurs forming the corresponding carboxylic acid
- When an alkene is heated under reflux with KMnO_4 , oxidation occurs and the $\text{C}=\text{C}$ double bond completely breaks, leaving a shorter chain carboxylic acid
- **Careful:** Alkenes will still undergo oxidation with cold dilute KMnO_4 , but the

conditions aren't as harsh so the $C=C$ will open up instead of completely breaking and a diol is formed

- This question is an excellent example of how the reactions of different organic species are often tested, so make sure you know these reactions thoroughly

3d

d) The rate of the formation of compound **W** is slower heating under reflux with aqueous silver nitrate rather than heating under reflux with aqueous sodium hydroxide because:

- The hydroxide ion is a better nucleophile (than water); [1 mark]
- As it has a full negative charge

AND

Water only has a partial negative charge; [1 mark]

[Total: 2 marks]

- Both reactions are nucleophilic substitution reactions whereby the bromine atom is replaced by the OH group forming butan-1-ol
- The rate of reaction depends on the ability of the nucleophile to donate its pair of electrons to the carbon atom bonded to Br
- The more negative the nucleophile, the more able it is to donate electron pairs

4a

a) An experiment that would enable the relative rates of hydrolysis of halogenoalkanes to be compared should include:

- Using ethanol; [1 mark]
- Using silver nitrate; [1 mark]
- Using equal amounts of each halogenoalkane; [1 mark]
- Measuring the time taken for the precipitate to form; [1 mark]
- Using a water bath (to control a raised temperature); [1 mark]

[Total: 5 marks]

- Your plan should include control variables such as equal amounts of the halogenoalkane
- Halogenoalkanes are insoluble in water, so using ethanol as a solvent allows water

and the halogenoalkane to mix

4b

b) The trend in the rates of hydrolysis of 1-chlorobutane, 1-bromobutane and 1-iodobutane is:

- The reaction rate is in the order: 1-chlorobutane < 1-bromobutane < 1-iodobutane; [1 mark]
- Because the C-Cl bond is stronger than the C-Br bond which is stronger than the C-I bond; [1 mark]

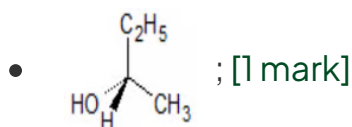
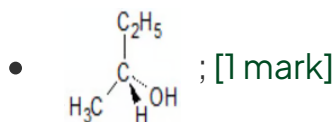
[Total: 2 marks]

- The second marking point is only awarded on the basis that your first point is correct
- If you give the wrong trend, you will score 0
- This is a common trend to ask about
 - A yellow precipitate of silver iodide would be observed forming first, as the C-I bond is the weakest
 - A cream precipitate of silver bromide would then be observed
 - A white precipitate of silver chloride would be observed forming last as the C-Cl bond is the hardest to break

4c

c) The term **chiral** means:

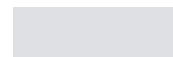
- It forms (two) isomers which are non-superimposable; [1 mark]



[Total: 3 marks]

- The diagrams should both be 3D, so must include 'wedges'
- You can score a standalone mark for the description of what chiral means even if your diagrams are absent or incorrect
- There are four different groups attached to the central carbon atom so the enantiomers are mirror images of each other and therefore non-superimposable (this description would also achieve marking point 1)

4d

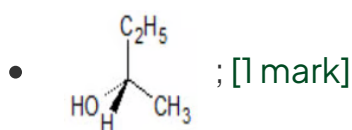
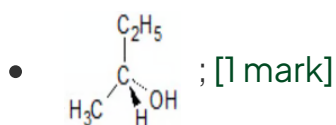


- A white precipitate of silver chloride would be observed forming last as the C-Cl bond is the hardest to break

4c

c) The term **chiral** means:

- It forms (two) isomers which are non-superimposable; [1 mark]



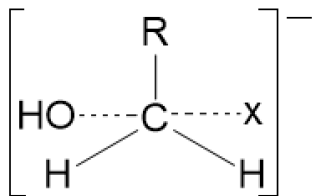
[Total: 3 marks]

- The diagrams should both be 3D, so must include 'wedges'
- You can score a standalone mark for the description of what chiral means even if your diagrams are absent or incorrect
- There are four different groups attached to the central carbon atom so the enantiomers are mirror images of each other and therefore non-superimposable (this description would also achieve marking point 1)

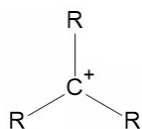
4d

d) Compare and contrast the mechanism of hydrolysis, using aqueous sodium hydroxide, of the primary halogenoalkane, RCH_2X , with that of the tertiary halogenoalkane, R_3CX :

- Both are nucleophilic substitution; [1 mark]
- Hydrolysis mechanism for RCH_2X / primary is $\text{S}_{\text{N}}2$ (via a transition state) **AND** R_3CX / tertiary is $\text{S}_{\text{N}}1$ (via a carbocation / intermediate); [1 mark]
- RCH_2X and OH^- in the rate-determining / slow step; [1 mark]
- R_3CX only in the rate-determining / slow step; [1 mark]
- (RCH_2X forms a transition state with OH^-) diagram, including dotted lines and charge: [1 mark]



- (R_3CX forms a carbocation / intermediate) diagram, including charge: [1 mark]



[Total: 6 marks]

- As the question asks you to compare, make sure that you give any similarities too
- The numbers in S_N1 and S_N2 refer to the number of species that the rate of reaction depends on i.e. how many species are in the rate-determining or slowest step
- Primary halogenoalkanes will react via a S_N2 mechanism which involves 1 step via a transition state
- Tertiary halogenoalkanes will react via a S_N1 mechanism which involves 2 steps via a carbocation
- **Note:** Secondary halogenoalkanes will undergo a mixture of these mechanisms, depending on their structure
- Make sure you give a diagram of these intermediates as prompted in the question
- It is important that you learn the mechanisms for each of these

5a

a) The structures of the four structural isomers of C_4H_9Br **and** their identities are:

- $CH_3CH_2CH_2CH_2Br$ primary; [1 mark]
- $CH_3CH_2CHBrCH_3$ secondary; [1 mark]
- $(CH_3)_2CHCH_2Br$ primary; [1 mark]
- $(CH_3)_3CBr$ tertiary; [1 mark]

[Total: 4 marks]

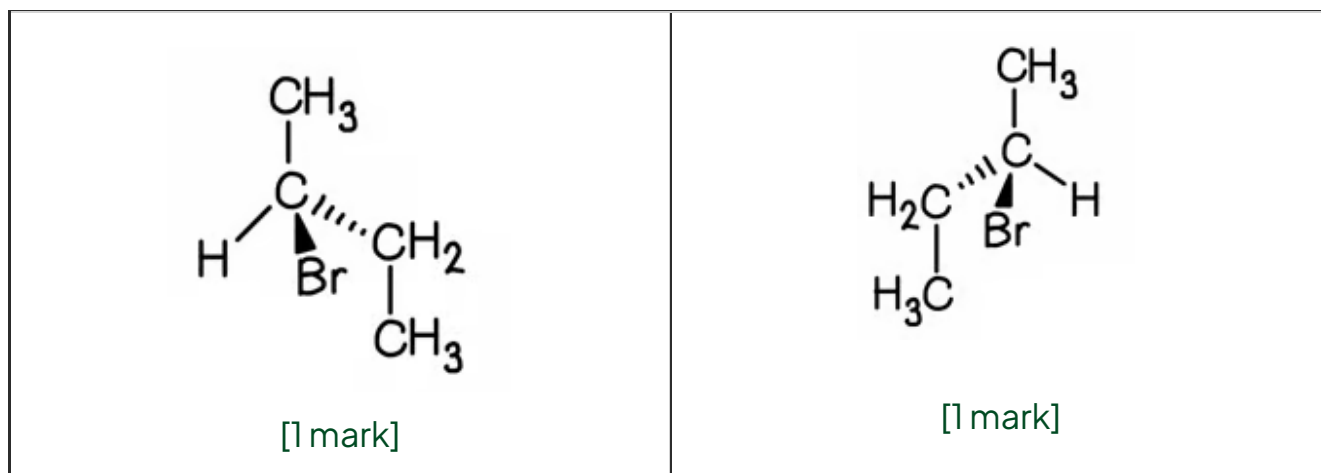
- When drawing isomers, take a systematic approach
- Draw the straight chain isomers first
 - Bromine can be on carbon 1 or 2

- Then identify the branched chain isomers
 - A methyl group can be on carbon 2 with bromine on carbon 1
 - 2 methyl groups can be on carbon 1 with bromine also on carbon 1
- **Remember:** To identify whether it is a primary, secondary or tertiary halogenoalkane, count how many alkyl groups the carbon which is attached to the halogen, is itself attached to

5b

b) The name of the isomer of C_4H_9Br that contains a chiral centre and the three-dimensional structures of the two optical isomers are:

- Name: 2-bromobutane; [1 mark]
- Structures:



[Total: 3 marks]

- A chiral center is a carbon atom that has four different atoms or groups of atoms attached
- In 2-bromobutane, carbon 2 is attached to H, Br, CH_3 and CH_2CH_3
- When drawing the optical isomers, make sure you draw wedges to show them in three dimensions

5c

c) The observations can be explained by:

- The halide ions released / liberated (from the hydrolysis of halogenoalkanes) for a precipitate with silver ions / Ag^+

OR



- The order is due to the decreasing bond strength ($\text{C-I} < \text{C-Br} < \text{C-Cl}$); [1 mark]

[Total: 2 marks]

- Silver ions will react with halide ions to form a precipitate
- Silver chloride will be the slowest precipitate to form as chloroalkanes are less reactive than bromoalkanes and iodoalkanes
- Silver iodide will be the quickest precipitate to form as iodoalkanes are the most reactive halogenoalkane
- This reactivity of the halogenoalkanes is determined by the strength of the C-X bond, the stronger the bond, the less reactive the halogenoalkane
- Therefore C-Cl bond is the strongest so will have the highest bond energy and C-I bond is the weakest and will have the lowest bond energy

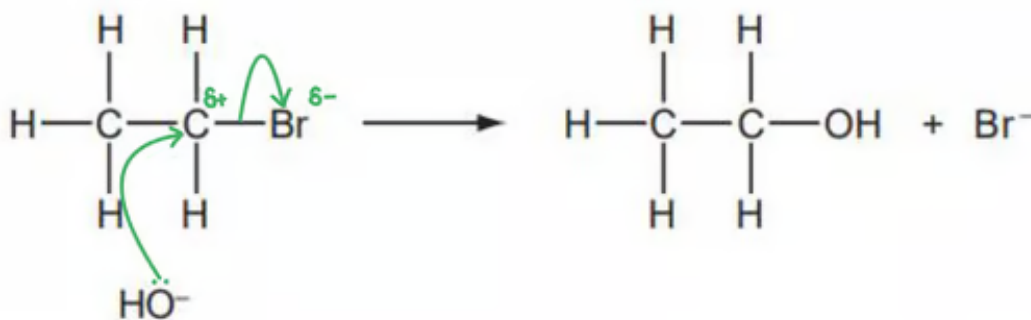
5d

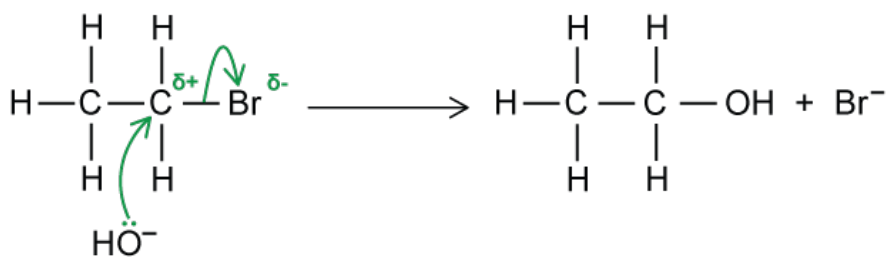
d)

i) The full name of the mechanism for the reaction between aqueous sodium hydroxide and bromoethane is:

- Nucleophilic; [1 mark]
- Substitution; [1 mark]

ii) The completed diagram to represent this mechanism is:





- Curly arrow from lone pair on OH^- ; [1 mark]
- Curly arrow from C-Br bond to Br **AND** dipole; [1 mark]

[Total: 4 marks]

- The OH^- ion acts as a nucleophile as it donates a pair of electrons to the δ^+ carbon atom bonded to Br
- Br is replaced by OH^- so it is a substitution reaction
- You should be able to draw this mechanism - remember to include any dipoles, lone pairs and any curly arrows should either start from a lone pair of electrons or a bond

5e

e)

i) The name that is given to the reaction mechanism when aqueous sodium hydroxide is reacted with primary or tertiary halogenoalkanes is:

- Primary halogenoalkane: $\text{S}_{\text{N}}2$; [1 mark]
- Tertiary halogenoalkane: $\text{S}_{\text{N}}1$; [1 mark]

ii) Tertiary carbocations are more stable than primary carbocations because:

- Alkyl groups have a positive inductive effect; [1 mark]
- In tertiary halogenoalkanes, there are 3 alkyl groups but in primary, there are one alkyl group; [1 mark]
- The charge density of the positive carbocation is reduced more by three alkyl groups / in tertiary halogenoalkane (than one alkyl group / in primary halogenoalkane); [1 mark]

[Total: 5 marks]

- **S_N** stands for nucleophilic substitution and the numbers refer to the number of species in the rate-determining step
 - **Remember:** The S_N2 mechanism is a one-step reaction whereas S_N1 occurs in two steps, via a carbocation intermediate
- Part (ii) asks you to explain why tertiary carbocations are more stable than primary carbocations, so you must refer to both in your answer: just referring to tertiary carbocations means you cannot score the final two marks